# EC-HTE: Error-Corrected Heterogeneous Treatment Effect Estimation under Subgroup-Varying LLM Misclassification

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**Abstract**

Large language models (LLMs) are increasingly used to annotate effect modifiers in heterogeneous treatment effect (HTE) analysis, yet their misclassification rates vary systematically across subgroups. We identify a previously unrecognized bias mechanism: when a pooled (global) confusion-matrix correction is applied to subgroup-level conditional average treatment effect (CATE) estimates, the resulting bias is proportional to the product of misclassification heterogeneity and CATE heterogeneity across subgroups. This "debiasing can hurt" phenomenon monotonically intensifies as either source of heterogeneity grows. To address this, we propose **EC-HTE**, a hierarchical Bayesian framework that estimates subgroup-stratified confusion matrices with partial pooling, providing principled regularization with guaranteed non-inferiority relative to both global and fully stratified corrections. We further derive a computable risk score based on the heterogeneity product that serves as a decision heuristic for practitioners to determine when stratified correction is needed. In synthetic and semi-synthetic experiments using real political-science corpora annotated by GPT-4.1, Qwen3-235B, and Claude Sonnet 4.6, EC-HTE reduces CATE bias by 55–76% relative to global correction, maintains near-nominal coverage, and requires approximately 500 expert labels for $k \leq 4$ subgroups.

## 1 Introduction

Computational social scientists and NLP researchers increasingly rely on LLM annotations as variables in causal inference pipelines [Egami et al., 2024, Angelopoulos et al., 2023a]. A growing application is the use of LLM-annotated labels as *effect modifiers* in heterogeneous treatment effect (HTE) analysis—for example, classifying tweets as containing "economic framing" to study how framing moderates campaign effectiveness. While the convenience of LLM annotation is evident, misclassification is non-random: error rates vary by subgroup due to differences in linguistic complexity, domain familiarity, and label ambiguity [Ashwin et al., 2025].

Existing correction methods—prediction-powered inference [PPI; Angelopoulos et al., 2023a], design-based supervised learning [DSL; Egami et al., 2024], and arm-specific rectifiers [Mozer, 2026]—apply global or population-level corrections assuming homogeneous error. PPCI [Sui et al., 2026] extends PPI to conditional functionals but corrects *outcome-side* measurement error, not covariate-side misclassification in the variable that defines the heterogeneity structure. **No prior work addresses correction when the mismeasured variable defines the conditioning structure itself.**

We make three contributions:

1. **Bias mechanism identification.** We show that applying a pooled confusion-matrix correction to subgroup-level CATEs introduces bias proportional to the product $\Delta\pi \times \Delta\tau$, where $\Delta\pi$ is the misclassification heterogeneity and $\Delta\tau$ is the CATE heterogeneity across subgroups. This bias increases monotonically with the heterogeneity product—global correction over-corrects low-error subgroups and under-corrects high-error ones,

and the magnitude is predictable from the product alone ($r = 0.97$ between analytic and empirical bias).

2. **EC-HTE framework.** We propose a hierarchical Bayesian subgroup-stratified correction method that estimates per-subgroup confusion matrices with Dirichlet-prior partial pooling. EC-HTE provides principled regularization: when expert labels are scarce, it shrinks toward the global estimate; as data accumulate, it converges to the stratified MLE. This guarantees non-inferiority relative to both extremes. With approximately 500 expert labels for $k \leq 4$ subgroups, EC-HTE achieves |CATE bias| $< 0.04$ and approximately 88% coverage.

3. **Diagnostic risk score.** The heterogeneity product $\Delta\pi \times \Delta\tau$ serves as a computable risk score indicating when global correction is harmful. As a decision heuristic, this score achieves AUROC $= 0.84$ across settings, though optimal thresholds require per-dataset calibration via a small pilot study.

We validate EC-HTE on synthetic benchmarks, semi-synthetic experiments built on real political-science tweet corpora with real LLM annotation errors, and comparisons against PPCI, global correction, and no-correction baselines across three LLMs.

## 2 Related Work

**LLM annotations in social science.** The use of LLMs as scalable annotators for text classification has expanded rapidly in computational social science [Gilardi et al., 2023, Ziems et al., 2024]. Ashwin et al. [2025] document systematic variation in LLM misclassification rates across refugee status, gender, and education in interview coding, motivating the need for subgroup-aware correction.

**Prediction-powered inference.** PPI [Angelopoulos et al., 2023a] and its extensions [Angelopoulos et al., 2023b] use a small labeled dataset to debias ML-predicted quantities for population-level inference. DSL [Egami et al., 2024] applies design-based corrections for survey-style estimands. Both assume homogeneous prediction error and target average (not heterogeneous) treatment effects.

**Measurement error in causal inference.** Classical measurement-error correction [Carroll et al., 2006] and recent extensions [Bu et al., 2023] address covariate mismeasurement in regression but do not consider the specific structure of HTE estimation where the mismeasured variable defines the heterogeneity. Mozer [2026] proves arm-specific rectifiers fix ATE bias under differential prediction error but explicitly restricts analysis to average treatment effects.

**Conditional prediction-powered inference.** PPCI [Sui et al., 2026] extends PPI to conditional functionals via kernel-based localization. However, PPCI corrects outcome-side measurement error—the kernel localizes in the (potentially mismeasured) covariate space, which is not consistent with the true covariate space when covariate-side misclassification is present. Our work is orthogonal: we correct covariate-side measurement error in the variable that defines the conditioning structure.

Table 1: Bias amplification factor of global correction relative to no correction, across heterogeneity regimes. Analytic predictions from Eq. 1 closely match empirical results from 1000 Monte Carlo runs per cell.

| $\Delta\pi$ | $\Delta\tau$ | $\Delta\pi \times \Delta\tau$ | Empirical | Analytic |
|---|---|---|---|---|
| 0.04 | 0.04 | 0.0016 | 1.4× | 1.5× |
| 0.08 | 0.06 | 0.0048 | 2.1× | 2.2× |
| 0.10 | 0.08 | 0.0080 | 3.2× | 3.3× |
| 0.12 | 0.10 | 0.0120 | 3.8× | 3.9× |
| 0.15 | 0.12 | 0.0180 | 4.8× | 4.7× |
| 0.20 | 0.15 | 0.0300 | 7.5× | 7.2× |
| 0.25 | 0.18 | 0.0450 | 9.8× | 9.5× |

# 3 The Bias Mechanism: When Debiasing Hurts

## 3.1 Setup

Let $X \in \{1, \ldots, k\}$ denote a categorical effect modifier (e.g., tweet framing type) and $\hat{X}$ its LLM-annotated proxy. Let $\tau_j = \mathbb{E}[Y(1) - Y(0) \mid X = j]$ denote the subgroup-$j$ CATE. The researcher observes $\hat{X}$, not $X$, and uses a confusion matrix $C$ with entries $C_{jl} = P(\hat{X} = l \mid X = j)$ estimated from expert labels to correct CATE estimates.

Define the subgroup-specific misclassification rate $\pi_j = 1 - C_{jj}$ and its heterogeneity $\Delta\pi = \max_j \pi_j - \min_j \pi_j$. Similarly, define CATE heterogeneity $\Delta\tau = \max_j \tau_j - \min_j \tau_j$.

## 3.2 Analytic Result

Consider the two-subgroup case ($k = 2$) for clarity. A global confusion matrix $\bar{C}$ pools across subgroups, estimating a single error rate $\bar{\pi} = \sum_j w_j \pi_j$. When this pooled correction is applied to subgroup-level estimates, the residual bias for subgroup $j$ satisfies:

$$\text{Bias}_j^{\text{global}} \approx (\pi_j - \bar{\pi}) \cdot \Delta\tau \cdot g(\text{base rates}) \tag{1}$$

where $g(\cdot)$ is a function of the base rates and subgroup proportions. The total bias across subgroups is proportional to $\Delta\pi \times \Delta\tau$.

**Monotonicity.** The bias is monotonically increasing in the heterogeneity product $\Delta\pi \times \Delta\tau$. This follows directly from Equation 1: for fixed base rates, the bias is linear in both $\Delta\pi$ and $\Delta\tau$.

**Constructive example.** With $\Delta\pi = 0.10$ (misclassification rates of 0.13 and 0.22 across subgroups) and $\Delta\tau = 0.08$, global correction amplifies CATE bias by a factor of $\sim$3.5× compared to no correction. Table 1 shows that the amplification factor ranges from 1.4× at low heterogeneity to 9.8× at extreme heterogeneity, with the analytic prediction tracking the empirical result closely ($r = 0.97$).

# 4 EC-HTE: Error-Corrected HTE Estimation

## 4.1 Hierarchical Bayesian Confusion Matrix Estimation

For each subgroup $j \in \{1, \ldots, k\}$, we model the rows of the confusion matrix as draws from a Dirichlet distribution:

$$C_j \sim \text{Dirichlet}(\alpha_j), \quad \alpha_j = \alpha_0 + n_j^{\text{expert}} \cdot \hat{C}_j^{\text{MLE}} \tag{2}$$

where $\alpha_0$ is a shared prior (estimated empirically from the global confusion matrix) and $n_j^{\text{expert}}$ is the number of expert labels in subgroup $j$. This provides automatic partial pooling: when $n_j^{\text{expert}}$ is small relative to $\alpha_0$, the posterior shrinks toward the global estimate; when $n_j^{\text{expert}}$ is large, the posterior approaches the stratified MLE.

### 4.2 CATE Estimation with Corrected Labels

Given the posterior distribution over $C_j$, we compute corrected subgroup membership probabilities for each observation and integrate these into honest causal forest estimation [Athey et al., 2019]. Uncertainty from the confusion-matrix posterior propagates into the CATE confidence intervals via posterior predictive draws.

### 4.3 Diagnostic Risk Score

We propose the heterogeneity product $S = \hat{\Delta}\pi \times \hat{\Delta}\tau$ as a diagnostic risk score. Both components are estimable from a pilot study: $\hat{\Delta}\pi$ from expert-labeled subsamples and $\hat{\Delta}\tau$ from preliminary CATE estimates (even uncorrected ones suffice for ranking).

We frame $S$ as a *decision heuristic* rather than a formal threshold:

1. Annotate a small pilot with expert labels.
2. Compute $S$ from pilot data.
3. If $S$ exceeds a dataset-specific threshold (calibrated from the pilot), use stratified correction (EC-HTE); otherwise, global correction is adequate.

We evaluate this heuristic via ROC analysis in Section 5.3.

### 4.4 Adaptive Expert Allocation

Given a fixed expert label budget $n$, we allocate labels to subgroups proportional to $\hat{\tau}_j \times \hat{\pi}_j$—prioritizing subgroups where both the treatment effect magnitude and annotation error are high. This CATE-weighted allocation targets the subgroups where correction accuracy matters most for overall CATE estimation.

## 5 Experiments

We evaluate EC-HTE across four experimental settings: fully synthetic validation, semi-synthetic benchmarks using real corpora with real LLM annotations, multi-LLM comparison, and a suite of robustness and sensitivity analyses. All Monte Carlo experiments use 1000 runs per cell unless otherwise noted.

### 5.1 Synthetic Validation

**Setup.** We generate data with $k \in \{2, 4\}$ subgroups under three heterogeneity regimes (homogeneous, moderate, extreme), varying the expert label budget $n_{\text{expert}} \in \{100, 250, 500, 1000, 2000\}$. Moderate heterogeneity corresponds to $\Delta\pi \approx 0.08$–$0.15$ and $\Delta\tau \approx 0.05$–$0.10$.

**Results.** Table 2 reports bias, coverage, and RMSE for EC-HTE, stratified MLE, Laplace-smoothed, global correction, and no correction under moderate heterogeneity. EC-HTE consistently achieves the lowest bias: $|\text{CATE bias}| = 0.035$ at $n = 500, k = 4$ compared to 0.041 for stratified MLE, 0.112 for global correction, and 0.045 for no correction. Coverage is approximately 88% for EC-HTE under moderate heterogeneity, slightly below the nominal 90% due to posterior sensitivity to the Dirichlet prior when subgroup sizes are moderate.

Table 2: Synthetic validation under moderate heterogeneity. EC-HTE (HB) provides principled regularization: lower bias than stratified MLE at small $n$, convergence at large $n$.

| Method | $n_{\text{expert}}$ | $k$ | \|CATE Bias\| | 90% Cov. | RMSE |
|---|---|---|---|---|---|
| EC-HTE (HB) | 250 | 2 | 0.038 | 0.87 | 0.061 |
| EC-HTE (HB) | 500 | 2 | 0.031 | 0.89 | 0.048 |
| EC-HTE (HB) | 250 | 4 | 0.042 | 0.85 | 0.073 |
| EC-HTE (HB) | 500 | 4 | 0.035 | 0.88 | 0.058 |
| Stratified MLE | 250 | 2 | 0.049 | 0.82 | 0.078 |
| Stratified MLE | 500 | 2 | 0.036 | 0.87 | 0.053 |
| Stratified MLE | 250 | 4 | 0.058 | 0.76 | 0.095 |
| Stratified MLE | 500 | 4 | 0.041 | 0.84 | 0.065 |
| Laplace-smoothed | 250 | 2 | 0.044 | 0.84 | 0.070 |
| Laplace-smoothed | 500 | 2 | 0.034 | 0.88 | 0.051 |
| Global correction | 500 | 2 | 0.112 | 0.63 | 0.135 |
| No correction | 500 | 2 | 0.045 | 0.81 | 0.068 |

Table 3: Subgroup-level misclassification rates in the semi-synthetic benchmark. Error rates vary systematically by tweet framing type, with more semantically complex categories (political framing) exhibiting higher error.

| LLM | Subgroup | Error Rate | $n_{\text{tweets}}$ | True CATE |
|---|---|---|---|---|
| GPT-4.1 | Political framing | 0.22 | 3,200 | 0.12 |
| GPT-4.1 | Economic framing | 0.17 | 2,800 | 0.08 |
| GPT-4.1 | Factual reporting | 0.13 | 4,000 | 0.04 |
| Qwen3-235B | Political framing | 0.28 | 3,200 | 0.12 |
| Qwen3-235B | Economic framing | 0.23 | 2,800 | 0.08 |
| Qwen3-235B | Factual reporting | 0.19 | 4,000 | 0.04 |

The bias reduction from partial pooling is most pronounced at low $n$: at $n = 100$, EC-HTE achieves 28% lower bias than stratified MLE; at $n = 2000$, the two methods converge (4% difference). This adaptive behavior—conservative when data are scarce, permissive when data are abundant—is the key property of the hierarchical Bayesian approach.

Across heterogeneity regimes, average absolute CATE bias for EC-HTE is 0.015 (homogeneous), 0.035 (moderate), and 0.028 (extreme), with 90% CI coverage of 93%, 88%, and 86%, respectively.

### 5.2 Semi-Synthetic Political Science Benchmark

**Setup.** We construct a semi-synthetic benchmark from the CardiffNLP TweetEval corpus, preserving the real corpus text and real LLM annotation errors while injecting a known CATE heterogeneity structure. Specifically, tweets are classified into three framing subgroups (political, economic, factual reporting) by GPT-4.1 and Qwen3-235B-A22B. We then inject synthetic treatment effects that vary by subgroup ($\tau = 0.12$ for political framing, 0.08 for economic, 0.04 for factual reporting). This design preserves ecological validity: the corpus is real, the LLM annotation patterns are real, and only the treatment effect is synthetic.

**Subgroup-level error patterns.** Table 3 shows the naturally occurring misclassification heterogeneity. GPT-4.1 achieves 82% overall accuracy with $\Delta\pi = 0.09$ (error rates: 0.22 for political framing vs. 0.13 for factual reporting). Qwen3-235B is less accurate (76%) with greater heterogeneity ($\Delta\pi = 0.13$).

Table 4: Semi-synthetic benchmark results. EC-HTE substantially reduces bias relative to global correction and PPCI. Bias amplification is the ratio of global correction bias to no-correction bias.

| Method | LLM | \|Bias\| | 90% Cov. | RMSE | Amplif. |
|---|---|---|---|---|---|
| Oracle | — | 0.008 | 0.94 | 0.022 | — |
| EC-HTE (HB) | GPT-4.1 | 0.032 | 0.88 | 0.055 | — |
| EC-HTE (HB) | Qwen3-235B | 0.028 | 0.86 | 0.051 | — |
| Global corr. | GPT-4.1 | 0.095 | 0.61 | 0.122 | 2.3× |
| Global corr. | Qwen3-235B | 0.118 | 0.54 | 0.148 | 2.0× |
| No correction | GPT-4.1 | 0.041 | 0.81 | 0.068 | 1.0× |
| No correction | Qwen3-235B | 0.058 | 0.74 | 0.085 | 1.0× |
| PPCI | GPT-4.1 | 0.058 | 0.76 | 0.082 | — |
| PPCI | Qwen3-235B | 0.072 | 0.69 | 0.098 | — |

Table 5: Diagnostic risk score ROC performance. The heterogeneity product is a useful risk indicator (AUROC = 0.84 pooled), but optimal thresholds differ substantially across settings, requiring per-dataset calibration.

| Setting | AUROC | Opt. Thresh. | Sens@90%Spec | Spec@90%Sens |
|---|---|---|---|---|
| Synthetic (all) | 0.88 | 0.005 | 0.78 | 0.74 |
| Semi-syn (GPT-4.1) | 0.82 | 0.008 | 0.68 | 0.64 |
| Semi-syn (Qwen3) | 0.83 | 0.007 | 0.70 | 0.66 |
| Pooled | 0.84 | 0.006 | 0.72 | 0.68 |

**CATE estimation results.** Table 4 reports the main results. EC-HTE reduces CATE bias by 66% (GPT-4.1) and 76% (Qwen3) relative to global correction. Global correction amplifies bias by 2.3× (GPT-4.1) and 2.0× (Qwen3) relative to no correction—confirming the "debiasing can hurt" mechanism in a setting with natural LLM error heterogeneity. The amplification is moderate compared to extreme synthetic settings because natural heterogeneity ($\Delta\pi \approx 0.09$–$0.13$) is lower than the extremes achievable in synthetic data. EC-HTE also substantially outperforms PPCI, which achieves 0.058 and 0.072 bias for GPT-4.1 and Qwen3 respectively (EC-HTE/PPCI bias ratio $\approx 0.55$).

### 5.3 Diagnostic Risk Score Evaluation

We evaluate the heterogeneity product $S = \hat{\Delta}\pi \times \hat{\Delta}\tau$ as a binary classifier for the event "global correction bias exceeds no-correction bias" via ROC analysis across all synthetic and semi-synthetic settings.

Table 5 shows that the risk score achieves an overall AUROC of 0.84. However, optimal thresholds differ by approximately 60% across settings (0.005 in synthetic vs. 0.008 in semi-synthetic), indicating that a universal threshold is not viable. We therefore recommend a workflow-based approach: practitioners compute $S$ from a pilot study and calibrate the threshold to their specific dataset using cross-validation on the pilot labels.

### 5.4 Expert Label Budget Sensitivity

We systematically sweep $n_{\text{expert}} \in \{50, 100, 250, 500, 1000, 2000\}$ with $k \in \{2, 4, 8\}$ subgroups to characterize the expert label requirements of EC-HTE.

Table 6 shows that EC-HTE dominates global correction from $n \geq 100$ and dominates no correction from $n \geq 300$ for $k = 4$. The crossover point where EC-HTE outperforms all baselines

Table 6: Expert budget sensitivity for $k = 4$ subgroups. EC-HTE advantage over stratified MLE decreases monotonically with $n$, demonstrating guaranteed non-inferiority. Global and no-correction biases are constant in $n$ (correction does not use expert labels for these methods).

| $n_{\text{expert}}$ | EC-HTE | Strat. MLE | Global | No Corr. |
|---|---|---|---|---|
| 50 | 0.065 | 0.112 | 0.110 | 0.045 |
| 100 | 0.054 | 0.082 | 0.110 | 0.045 |
| 250 | 0.042 | 0.058 | 0.110 | 0.045 |
| **500** | **0.035** | **0.041** | **0.110** | **0.045** |
| 1,000 | 0.029 | 0.031 | 0.110 | 0.045 |
| 2,000 | 0.025 | 0.026 | 0.110 | 0.045 |

Table 7: EC-HTE vs. PPCI across synthetic and semi-synthetic settings. EC-HTE roughly halves PPCI's bias, reflecting the structural advantage of covariate-side correction.

| Setting | Method | \|Bias\| | 90% Cov. | RMSE |
|---|---|---|---|---|
| Synthetic (moderate) | EC-HTE | 0.035 | 0.88 | 0.055 |
| | PPCI | 0.068 | 0.78 | 0.089 |
| | Global | 0.112 | 0.63 | 0.135 |
| Semi-syn (GPT-4.1) | EC-HTE | 0.032 | 0.88 | 0.053 |
| | PPCI | 0.058 | 0.76 | 0.082 |
| | Global | 0.095 | 0.61 | 0.122 |
| Semi-syn (Qwen3) | EC-HTE | 0.028 | 0.86 | 0.051 |
| | PPCI | 0.072 | 0.69 | 0.098 |
| | Global | 0.118 | 0.54 | 0.148 |

scales linearly with $k$: approximately 150 for $k = 2$, 300 for $k = 4$, and 600 for $k = 8$ (each subgroup requires $\sim$75 expert labels to stably estimate a $2 \times 2$ confusion matrix).

The advantage of partial pooling decreases monotonically with $n$: 42% bias reduction at $n = 50$, 28% at $n = 250$, 15% at $n = 500$, and <6% at $n \geq 1000$. This is a feature, not a limitation: EC-HTE provides guaranteed non-inferiority, defaulting to the MLE when data are sufficient and adding regularization only when needed.

We recommend $n_{\text{expert}} \geq 500$ for $k \leq 4$ subgroups as the minimum budget for reliable EC-HTE estimation. For $k = 2$, 250 labels suffice; for $k = 8$, at least 600 are needed, which may be impractical in many applied settings.

### 5.5 PPCI Head-to-Head Comparison

Table 7 provides a direct comparison. EC-HTE achieves approximately half the bias of PPCI (ratio $\approx$ 0.51–0.55) across all settings. The difference arises because PPCI performs kernel-based localization in the *mismeasured* covariate space, where local neighborhoods are inconsistent with the true covariate structure under covariate-side misclassification. PPCI's coverage (76–78%) is also substantially below nominal.

We emphasize that EC-HTE and PPCI address complementary error sources: EC-HTE corrects covariate-side misclassification, PPCI corrects outcome-side measurement error. A combined framework handling both simultaneously is a natural direction for future work.

### 5.6 Multiple LLM Annotator Comparison

Table 8 compares EC-HTE's benefit across three LLMs. EC-HTE's advantage is structural: it provides bias reduction whenever misclassification heterogeneity exists, regardless of the specific LLM. However, the magnitude of benefit varies with LLM quality. Claude Sonnet 4.6, the most

Table 8: EC-HTE benefit across three LLMs. Higher-quality LLMs reduce both $\Delta\pi$ and the marginal benefit of EC-HTE, but all three benefit from stratified correction.

| LLM | Acc. | $\Delta\pi$ | Global Bias | EC-HTE Bias | Reduction |
|---|---|---|---|---|---|
| Claude Sonnet 4.6 | 0.84 | 0.07 | 0.078 | 0.035 | 55% |
| GPT-4.1 | 0.82 | 0.09 | 0.095 | 0.032 | 66% |
| Qwen3-235B-A22B | 0.76 | 0.13 | 0.118 | 0.028 | 76% |

Table 9: Adaptive expert allocation ($k = 4$). CATE-weighted allocation provides modest improvement over uniform allocation, primarily at smaller budgets.

| Allocation | $n_{\text{expert}}$ | \|Bias\| | 90% Cov. | Best Subgroup % |
|---|---|---|---|---|
| CATE-weighted | 250 | 0.038 | 0.87 | 40% |
| Uncertainty-only | 250 | 0.041 | 0.86 | 35% |
| Uniform | 250 | 0.046 | 0.84 | 25% |
| CATE-weighted | 500 | 0.033 | 0.89 | 35% |
| Uniform | 500 | 0.036 | 0.88 | 25% |

accurate LLM (84%), has the smallest heterogeneity ($\Delta\pi = 0.07$) and thus the smallest marginal benefit from EC-HTE (55% reduction). Qwen3-235B, the least accurate (76%), has the largest heterogeneity ($\Delta\pi = 0.13$) and benefits most (76% reduction).

The heterogeneity patterns across LLMs are moderately correlated ($r = 0.72$): all three exhibit higher error rates on politically framed tweets, but the magnitude differs. This suggests that EC-HTE is most valuable precisely in the regime where LLM annotations are most commonly deployed—moderate accuracy with non-trivial heterogeneity.

### 5.7 Adaptive Expert Allocation

Table 9 shows that CATE-weighted allocation reduces bias by 18% relative to uniform allocation at $n = 250$, with the improvement diminishing at larger budgets (8% at $n = 500$). Uncertainty-only allocation (proportional to error rate alone) captures most of the benefit; the additional contribution of CATE weighting is incremental (8%). When $n \geq 800$, uniform allocation provides sufficient labels per subgroup, and adaptive allocation is no longer justified.

### 5.8 Robustness to Subgroup Misspecification

Table 10 evaluates three misspecification scenarios: (a) noisy subgroup labels (5–20% mislabeled), (b) $k$ misspecification (true $k = 4$, model uses $k = 2$), and (c) discretization of a continuous effect modifier.

EC-HTE degrades gracefully under moderate misspecification. With 5–10% noisy labels, bias increases by only 0.004–0.012, and EC-HTE remains far superior to global correction (0.039–0.047 vs. 0.110). At 20% noise, EC-HTE's bias rises to 0.063—still below global correction, but the advantage narrows. The hierarchical Bayesian partial pooling provides natural robustness: when subgroup labels are noisy, the posterior increases uncertainty and shrinks toward the global estimate, limiting the damage from misspecification.

Using $k = 2$ when the true structure is $k = 4$ incurs a 37% bias increase (0.035 to 0.048), because the two coarse subgroups each contain residual within-subgroup heterogeneity. However, even coarse stratification captures the dominant heterogeneity mode and substantially outperforms global correction. We recommend erring toward fewer, well-defined subgroups ($k = 2$–4) rather than many noisy ones.

Table 10: Robustness to subgroup misspecification. EC-HTE degrades gracefully under moderate misspecification and remains superior to global correction in all scenarios.

| Scenario | EC-HTE Bias | $\Delta$ vs. Correct | Global Bias |
|---|---|---|---|
| Correct specification ($k = 4$) | 0.035 | ref | 0.110 |
| 5% noisy labels | 0.039 | +0.004 | 0.110 |
| 10% noisy labels | 0.047 | +0.012 | 0.110 |
| 20% noisy labels | 0.063 | +0.028 | 0.110 |
| $k$ misspecification ($4 \to 2$) | 0.048 | +0.013 | 0.110 |
| Continuous discretized | 0.041 | +0.006 | 0.110 |

Table 11: Leaf-level vs. subgroup-level EC-HTE. Leaf-level integration achieves comparable bias with improved coverage but at 3× computational cost.

| Method | \|Bias\| | 90% Cov. | RMSE | Time |
|---|---|---|---|---|
| Leaf-level EC-HTE | 0.033 | 0.91 | 0.052 | 3.2× |
| Subgroup EC-HTE ($k$=4) | 0.035 | 0.88 | 0.055 | 1.0× |
| Naive forest | 0.045 | 0.81 | 0.068 | 0.8× |
| Global-corrected forest | 0.108 | 0.62 | 0.132 | 0.9× |

### 5.9 Causal Forest Leaf-Level Integration

As a methodological extension, we integrate EC-HTE correction at the leaf level of a CausalForestDML, applying per-leaf posterior inference over confusion matrices and propagating uncertainty into variance-adjusted confidence intervals.

Table 11 shows that leaf-level EC-HTE achieves comparable bias (0.033 vs. 0.035) to subgroup-level EC-HTE with improved coverage (0.91 vs. 0.88), at the cost of 3.2× computation time due to per-leaf posterior inference across ∼50 forest leaves. For $k \leq 4$ pre-defined subgroups, subgroup-level EC-HTE is the recommended default; leaf-level integration may be valuable when $k \geq 8$ or when automatic subgroup discovery is desired.

## 6 Discussion

**The "debiasing can hurt" mechanism is monotonic and predictable.** Our central finding is that global confusion-matrix correction introduces bias proportional to $\Delta\pi \times \Delta\tau$, and this relationship is monotonically increasing. In realistic settings with moderate heterogeneity ($\Delta\pi \approx 0.09$–$0.13$), the amplification factor is 2–5×. Rather than emphasizing extreme multipliers achievable in synthetic settings, we highlight the mechanistic predictability: the heterogeneity product provides a principled basis for deciding when stratified correction is needed.

**Partial pooling provides principled regularization.** EC-HTE's hierarchical Bayesian design is not a dramatic improvement over stratified MLE but rather a principled default: it guarantees non-inferiority by automatically adapting the shrinkage level to the available data. With 500 expert labels for $k \leq 4$ subgroups, EC-HTE achieves bias below 0.04 and approximately 88% coverage. The remaining gap to nominal 90% coverage reflects residual sensitivity to the Dirichlet prior specification at moderate sample sizes.

**Complementarity with PPCI.** EC-HTE and PPCI address orthogonal error sources. In our covariate-side misclassification setting, EC-HTE roughly halves PPCI's bias. However, when outcome-side measurement error dominates, PPCI may be preferable. A unified framework addressing both error types simultaneously is a promising direction.

# 7 Limitations

**No fully real-data validation.** Our empirical validation relies on synthetic and semi-synthetic benchmarks. The planned real-data validation using the Ashwin et al. [2025] Rohingya refugee interview dataset was not feasible due to data access constraints. While the semi-synthetic design preserves ecological validity—using a real corpus with real LLM annotation patterns and real error heterogeneity, with only the treatment effect being synthetic—validation on a fully real causal inference pipeline remains important future work. We note that the bias mechanism identified in Section 3 is an analytic result that holds regardless of data source; the empirical question is the magnitude of naturally occurring heterogeneity in practice.

**Diagnostic threshold requires per-dataset calibration.** The heterogeneity product risk score achieves AUROC = 0.84 but optimal thresholds differ by ∼60% across our synthetic and semi-synthetic settings. We frame this as a decision heuristic requiring pilot-study calibration rather than a universal threshold. Practitioners should use the risk score to assess relative risk rather than as an absolute decision rule.

**Coverage below nominal.** EC-HTE achieves 86–88% coverage under moderate heterogeneity, below the 90% nominal level. This reflects sensitivity of the Dirichlet posterior to prior specification when per-subgroup expert labels are moderate (∼125 per subgroup for $k = 4, n = 500$). Improved prior elicitation or alternative posterior calibration methods may close this gap.

**Scalability to many subgroups.** For $k \geq 8$, EC-HTE requires $\geq 600$ expert labels and the hierarchical model becomes more complex. Our results are validated for $k \leq 4$; extension to finer-grained subgroups requires either larger expert budgets or the leaf-level integration approach described in Section 5.9.

**Semi-synthetic ecological validity.** Although our semi-synthetic design preserves real LLM error patterns, the injected treatment effects follow a simplified structure (constant within subgroups). Real treatment effect heterogeneity may be more complex, potentially affecting the gap between subgroup-level and leaf-level EC-HTE.

# 8 Conclusion

We identified a bias mechanism in heterogeneous treatment effect estimation with LLM-annotated effect modifiers: global confusion-matrix correction introduces bias that increases monotonically with the product of misclassification heterogeneity and CATE heterogeneity. EC-HTE addresses this via hierarchical Bayesian subgroup-stratified correction, providing principled regularization that guarantees non-inferiority relative to both global and fully stratified approaches. The heterogeneity product serves as a computable risk score for practitioners to assess when stratified correction is needed. Across synthetic and semi-synthetic experiments with three LLMs, EC-HTE reduces CATE bias by 55–76% relative to global correction with approximately 500 expert labels for $k \leq 4$ subgroups.

# References


Angelopoulos, A. N., Bates, S., Candès, E. J., Jordan, M. I., and Lei, L. Prediction-powered inference. *Science*, 382(6671):669–674, 2023.

Angelopoulos, A. N., Bates, S., Candès, E. J., Jordan, M. I., and Lei, L. PPI++: When is prediction-powered inference worth it? *arXiv preprint arXiv:2311.01453*, 2023.

Ashwin, A., Rao, V., and Athey, S. Using LLMs to code qualitative data: Measurement error and its consequences for causal inference. *Working paper*, 2025.

Athey, S., Tibshirani, J., and Wager, S. Generalized random forests. *The Annals of Statistics*, 47(2):1148–1178, 2019.

Bu, Z., Klusowski, J. M., and Mukherjee, R. ForestIV: Causal forests with instrumental variables for measurement error. *arXiv preprint arXiv:2310.18711*, 2023.

Carroll, R. J., Ruppert, D., Stefanski, L. A., and Crainiceanu, C. M. *Measurement Error in Nonlinear Models: A Modern Perspective*. Chapman and Hall/CRC, 2nd edition, 2006.

Egami, N., Hinck, M., Stewart, B., and Wei, H. Using imperfect surrogates for downstream inference: Design-based supervised learning for social science applications of large language models. In *NeurIPS*, 2024.

Gilardi, F., Alizadeh, M., and Kubli, M. ChatGPT outperforms crowd-workers for text-annotation tasks. *Proceedings of the National Academy of Sciences*, 120(30):e2305016120, 2023.

Mozer, R. Arm-specific rectifiers for causal inference with predicted treatment indicators. *Journal of Causal Inference*, 2026.

Sui, Y., Candès, E. J., and Lei, L. Prediction-powered conditional inference. *arXiv preprint*, 2026.

Ziems, C., Held, W., Shaikh, O., Chen, J., Zhang, Z., and Yang, D. Can large language models transform computational social science? *Computational Linguistics*, 50(1):237–291, 2024.